CONFIDENTIAL

Development Environment Building (For Customer Use)

2026-9

Revision History

No.	Date	Note	version
01	2026.9	Initial release.	V1.0.0

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1 Introduction

This document is used to explain the construction of the development environment for the device, application compilation, firmware download, and program debugging.

2 Installation Of Compilation Environment

Unzip "*prebuilds.rar*" to the root directory of drive C. ATTENTION! After unzipping, the *prebuilds* directory can only be the direct subdirectory of root directory, as shown in Figure 2.1 below:

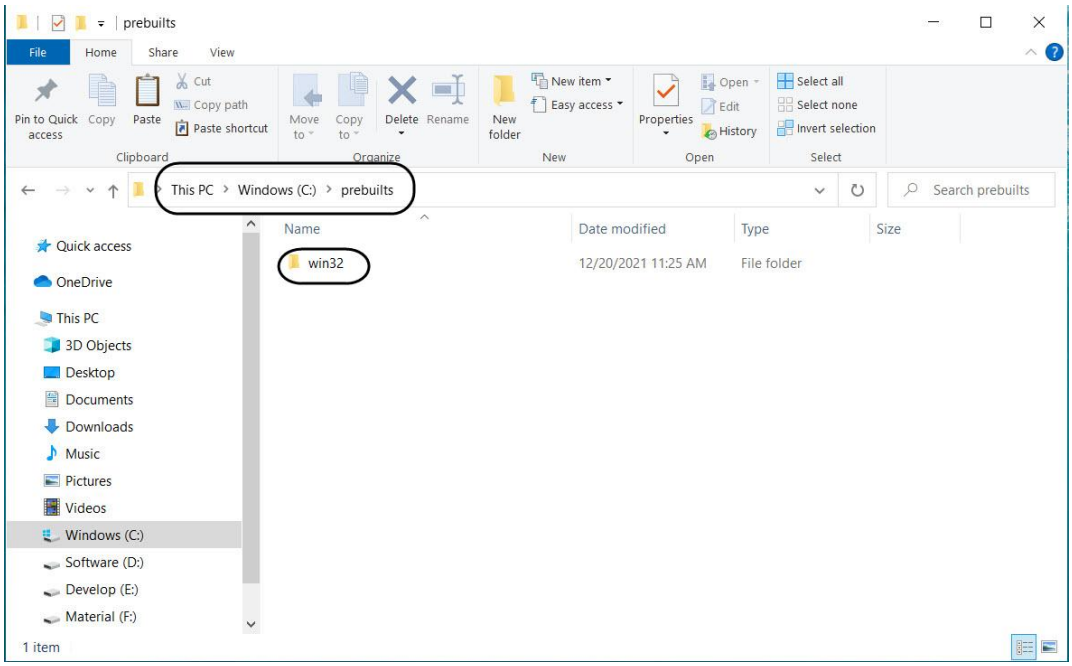


Figure 2.1 compilation tool directory

3 Debugging Board

As shown in Figure 3.1 below, the multi-functional debugging board integrates download and debugging functions, one USB 1-to-2 virtual serial port, and one USB download interface. The debugging board has two control switches: the BOOT download mode switch and the power control switch. Meanwhile, the board is equipped with a BOOT status LED, a power status LED, a USB communication port LED, and a USB virtual serial port emulation status LED. The on-board Type-C port is used to connect to the USB port of the device, and the two Micro USB ports are used to connect to the USB ports of the PC.

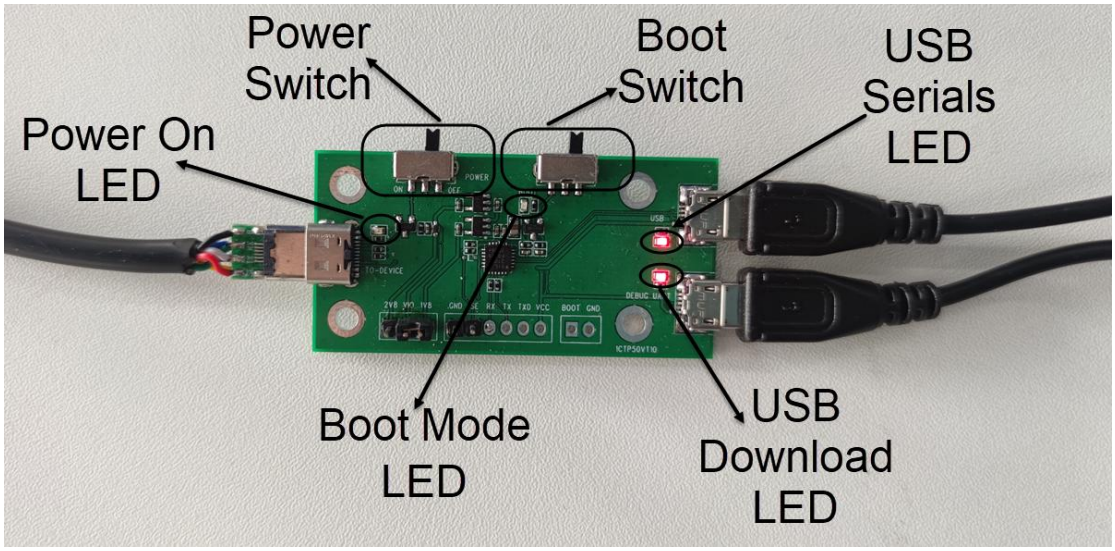


Figure 3.1 debugging board

Flip the BOOT switch to the left and the power switch to the "ON" end (left), and the BOOT Mode LED lights up. Connecting the Type-C port of the debugging board to the USB port of the device that is powered off makes the device enter the download mode. As shown in Figure 3.2, flip the BOOT switch to the right to exit BOOT mode, and then the USB port can start communication.

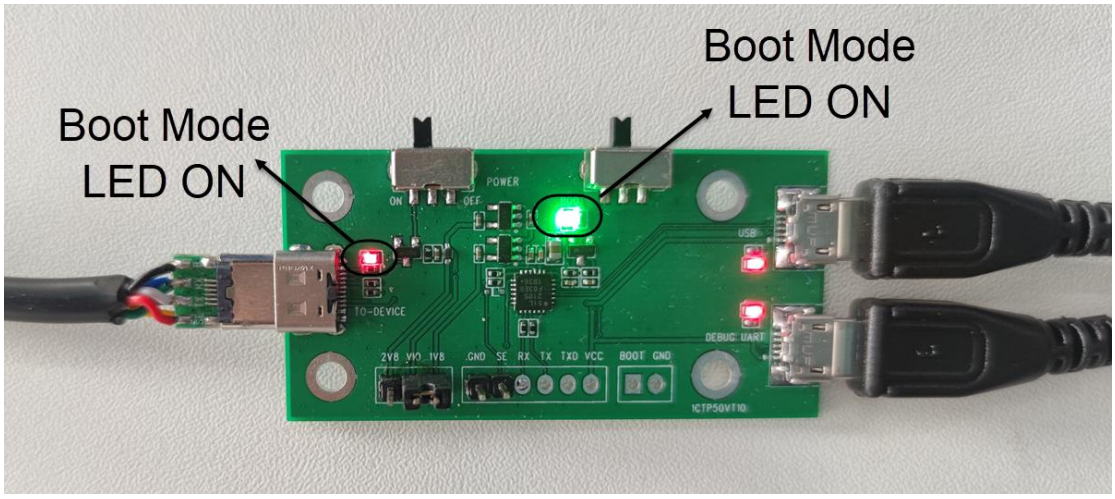


Figure 3.2 BOOT mode

4 Drive Installation

When the device is turned off, use two Micro USB cables to connect the debugging board to the USB ports of the development PC, and use a double-ended Type-C cable to connect the debugging board to the Type-C port of the device, as shown in Figure 4.1.

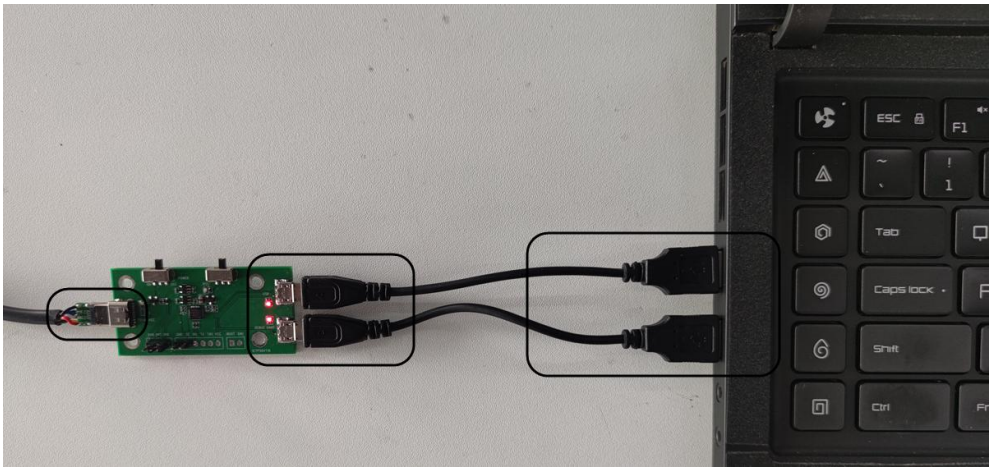


Figure 4.1 USB boot-A terminal connection

Unzip the "*8910_module_usb_driver_20191011_signed.7z*" drive package to the PC disk (the following steps vary according to the specific storage location), and record the storage path of the driver file. Following the steps shown in figure 4.2/4.3/4.4, select the drive directory to complete the drive upgrade.

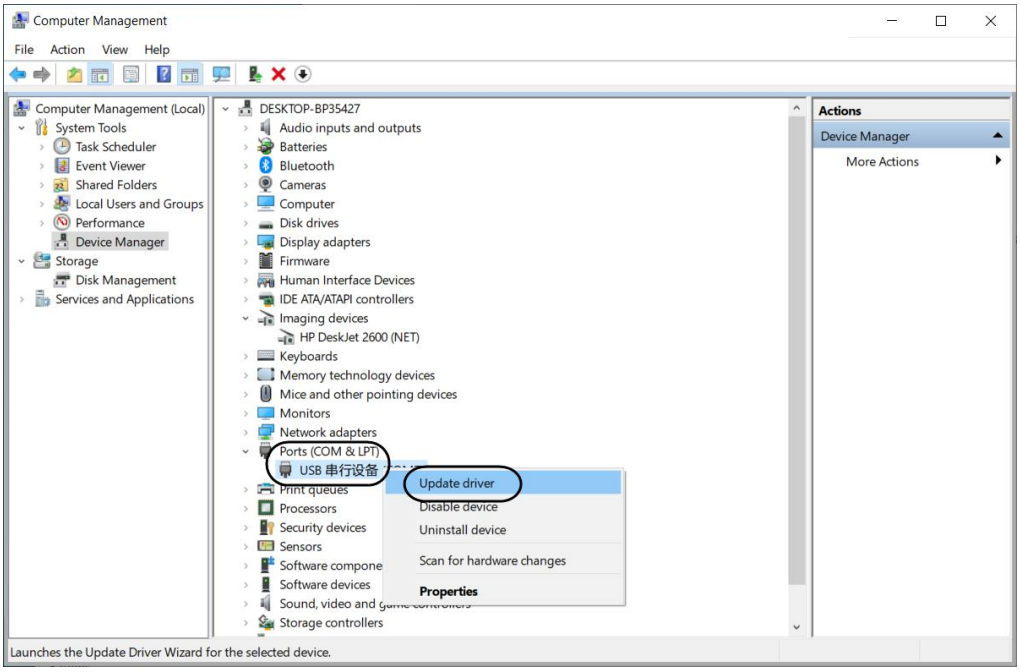


Figure 4.2 device unidentified

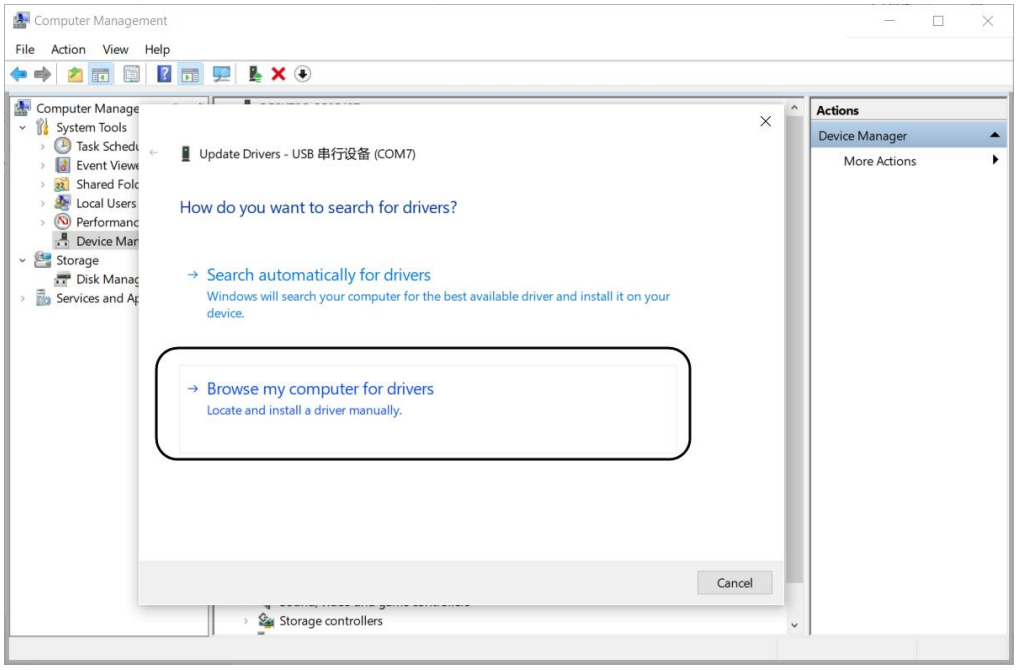


Figure 4.3 select manual search drive mode

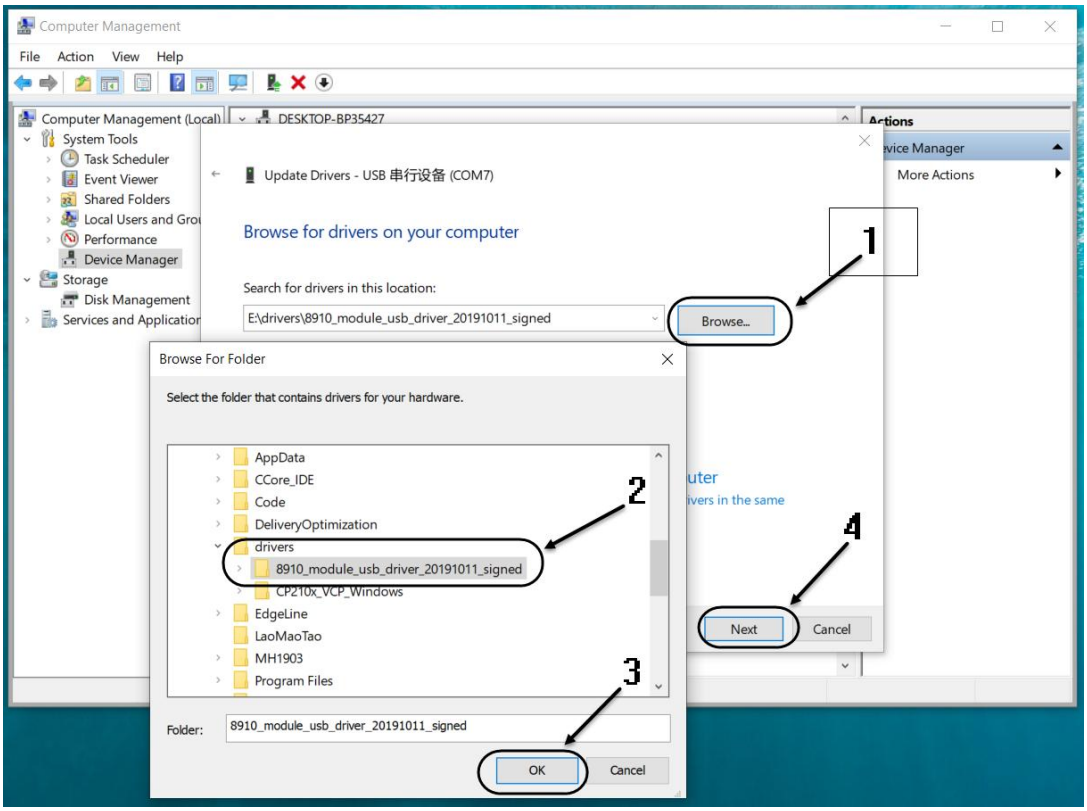


Figure 4.4 select drive-located directory

After installation of the drive, click "close" to complete the drive installation, as shown in figure 4.5 below.

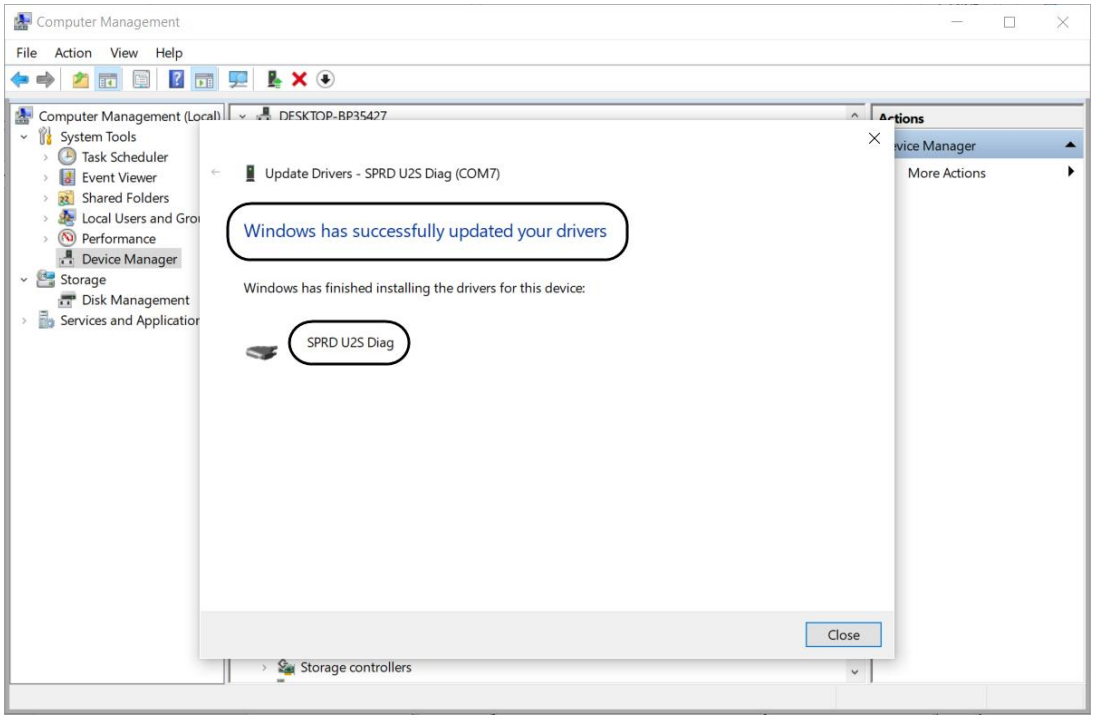


Figure 4.5 driver installation complete

5 Application compilation

Unzip the "app_csdk.rar" package to get the source code project of the SDK.

execute "xx_debug_build.bat" to complete the compilation. firmware file will be generated in the "\out\xx_debug\hex" directory. This firmware can output application logs.

execute "xx_release_build.bat" to complete the compilation. firmware file will be generated in the "\out\xx_release\hex" directory. This firmware cannot output application logs.

The method of use is as follows：

In the directory where the build script is located, open a command line window (Figures 5.1 and 5.2):

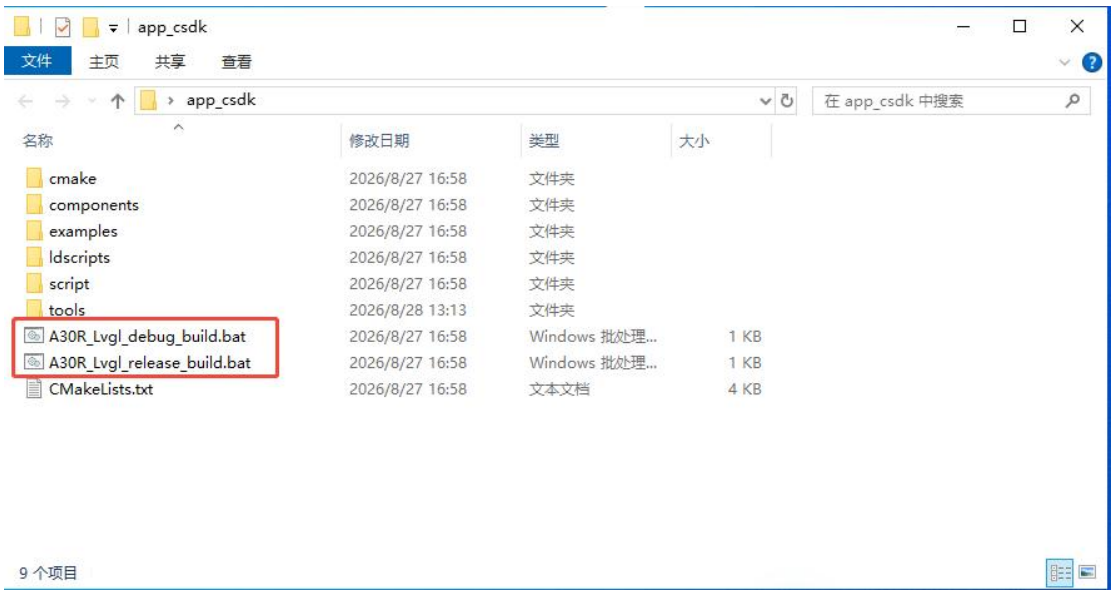


Figure 5.1 File Directory

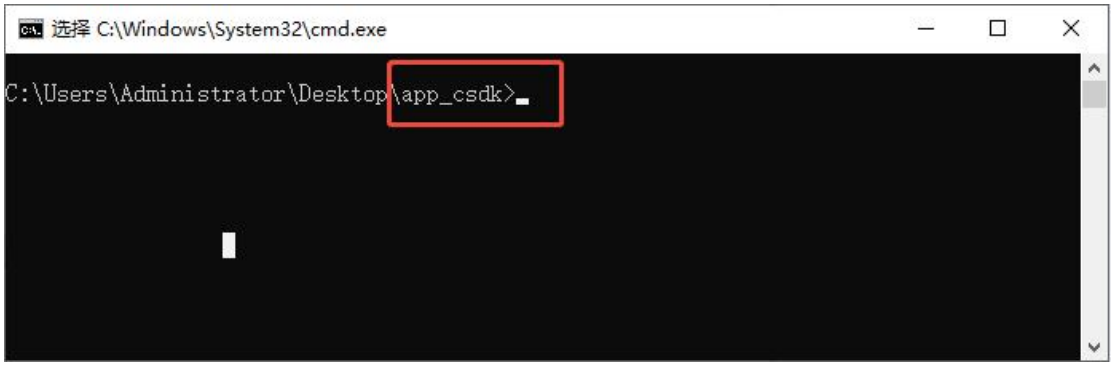


Figure 5.2 Windows command line window

2. Enter the compilation script and the name of the compilation log file, such as build_log.txt, and enter the command as follows:

xx_release_build.bat >build_log.txt(Figure 5.3)

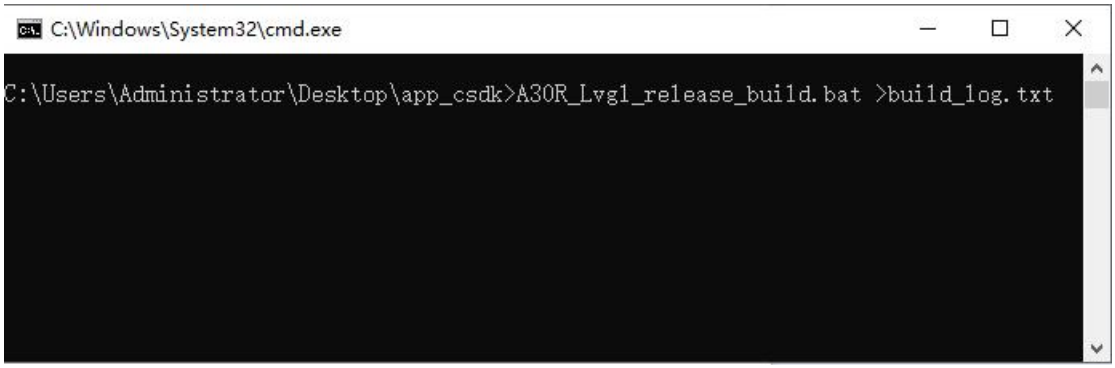


Figure 5.3 Enter the compile command

3.After the compilation is complete, generate the build_log.txt file(Figure 5.4):

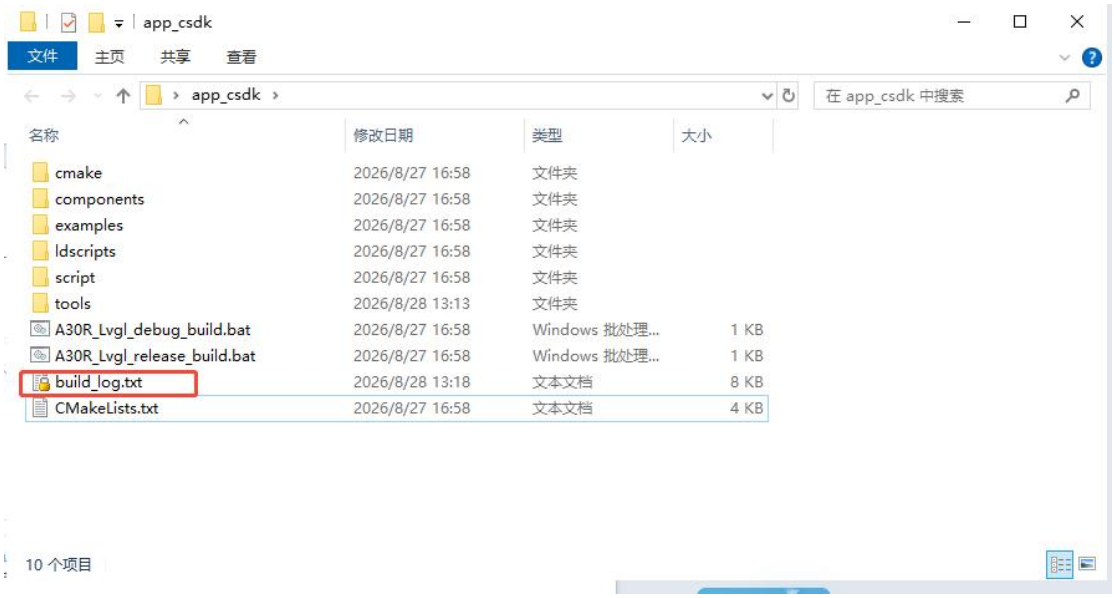


Figure 5.4 Compile log file

4. View compilation results:

4.1 Compiled successfully(Figure 5.5):

```
[35/36] Linking C executable hex\mainapp_flash.elf
[36/36] Generating ../../hex/mainapp_flash.pac
        1 file(s) copied.
        1 file(s) copied.
```

Figure 5.5 Compiled successfully

4.2 Compilation fails(Figure 5.6):

In the build_log.txt file, you can search for the "error" string and analyze the cause of the compilation failure.

```
ninja: build stopped: subcommand failed.
```

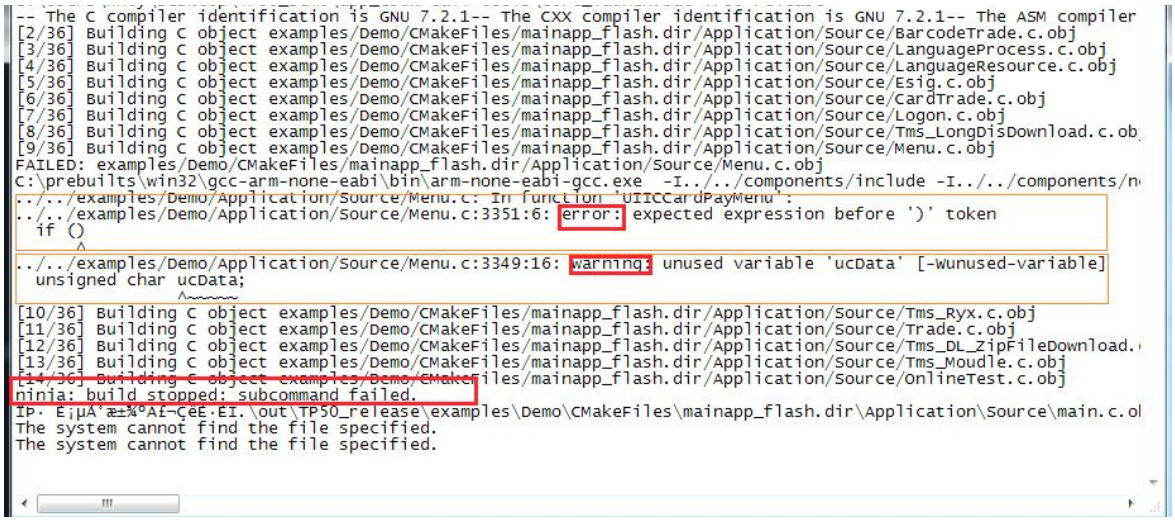


Figure 5.6 Compilation fails

6 PAC Firmware Downloads

PAC firmware download is suitable for updating PAC files generated by OS and application packaging, mainly used for new device production, OS firmware version update and recovery after device firmware damage.

When the device is turned off, use the Type C Boot cable to connect between the USB ports of the PC and the device, as described in Chapter 3. Open the "UpgradeDownload.exe" tool, use the "Load packet" button, and select the "XX _Vx.00.00_Rxxxx_sign_APP_Vx.00.00.Rxxxx.pac" File, as shown in Figure 6.1 below. Then click "Start downloading" (figure 6.2)to start updating the device firmware process(figure 6.3). After the download and installation is complete, the green word “passed” will indicate completion(figure 6.4).

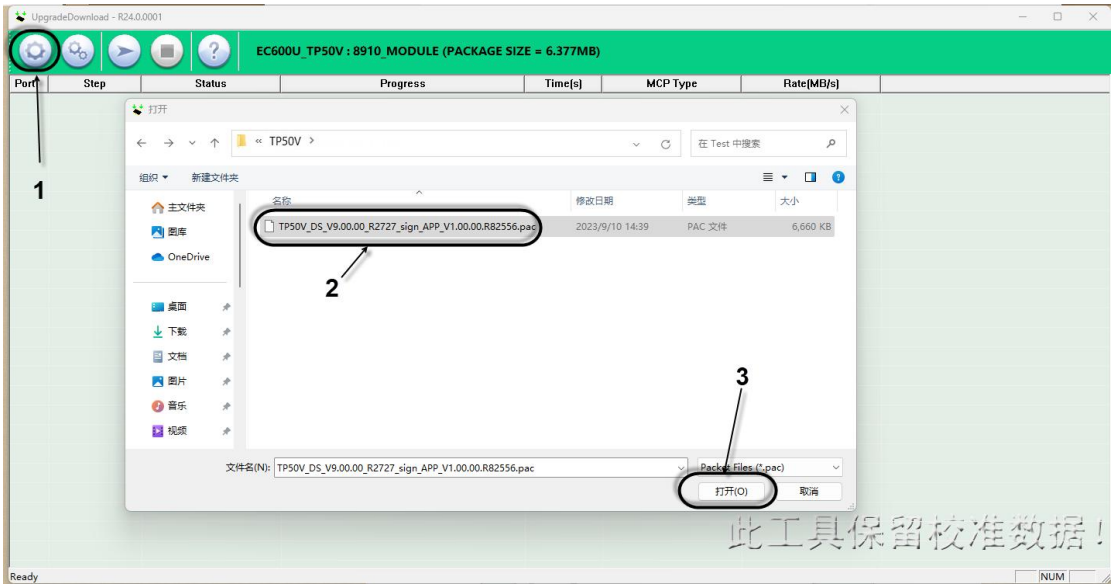


Figure 6.1 choose the pac file



Figure 6.2 start download

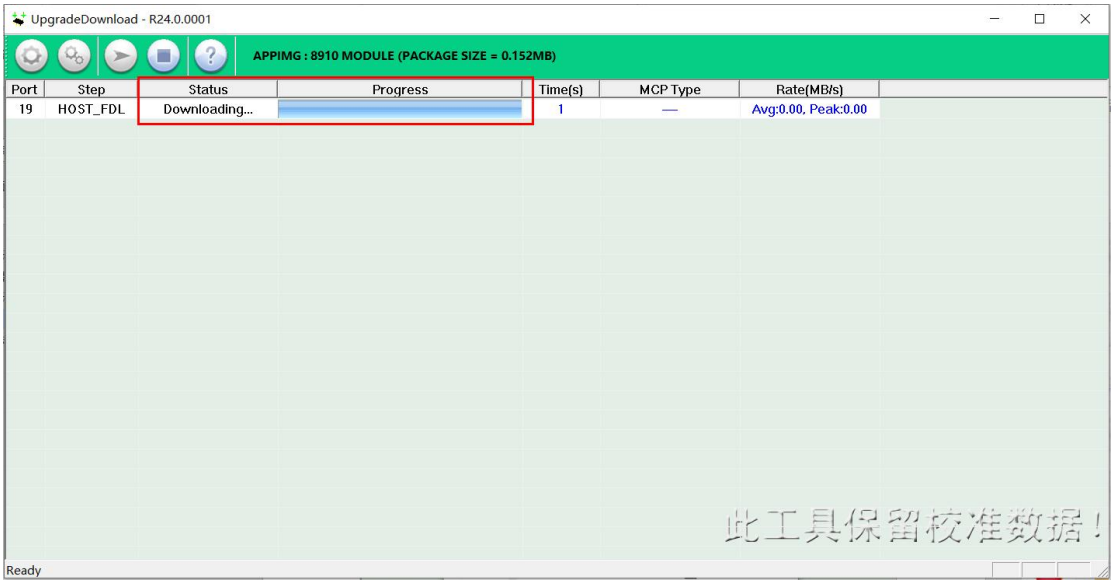


Figure 6.3 firmware downloading

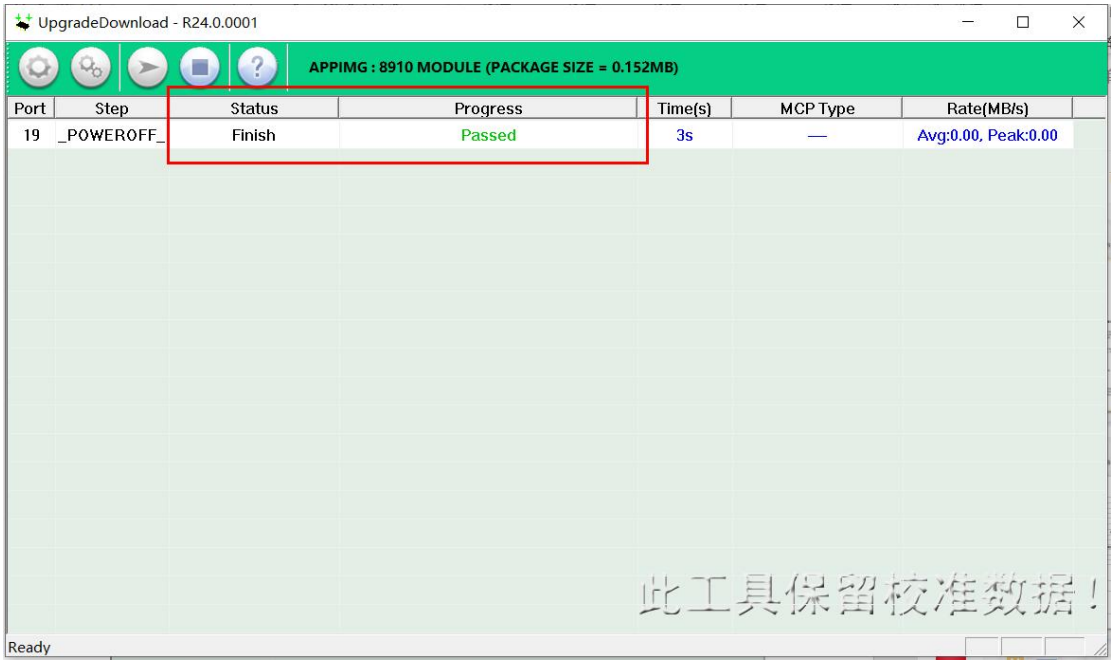


Figure 6.4 firmware download complete

7 Application firmware download

Application update is mainly used to update the device application firmware in the debugging phase.

The device first enters Maintenance mode and the operation steps are as follows:Main Menu->More->Maintenance. Open the ‘AppUpgrade Tool.exe' tool and click on the 'Device' drop-down box，Select ‘CCID READER'，'File' Select the. img file compiled by the DemoApp, Click ‘Download’ to start downloading (figure 7.1).

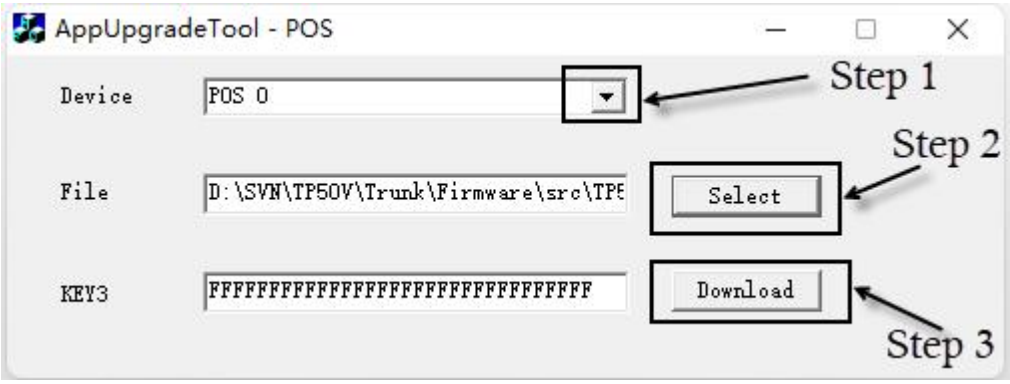


Figure 7.1 Application firmware download

Note: KEY3 cannot be modified.

8 Debugging

8.1 Cable connection

Use two Micro USB cables to connect the PC USB ports to the debugging board(figure 8.1). Use the DuPont cable to insert the serial port module pins, and connect the DuPont cable to the Debug serial port pins of the device.

Unzip " CP210x_VCP_Windows.rar" to the PC disk, and record the save directory.

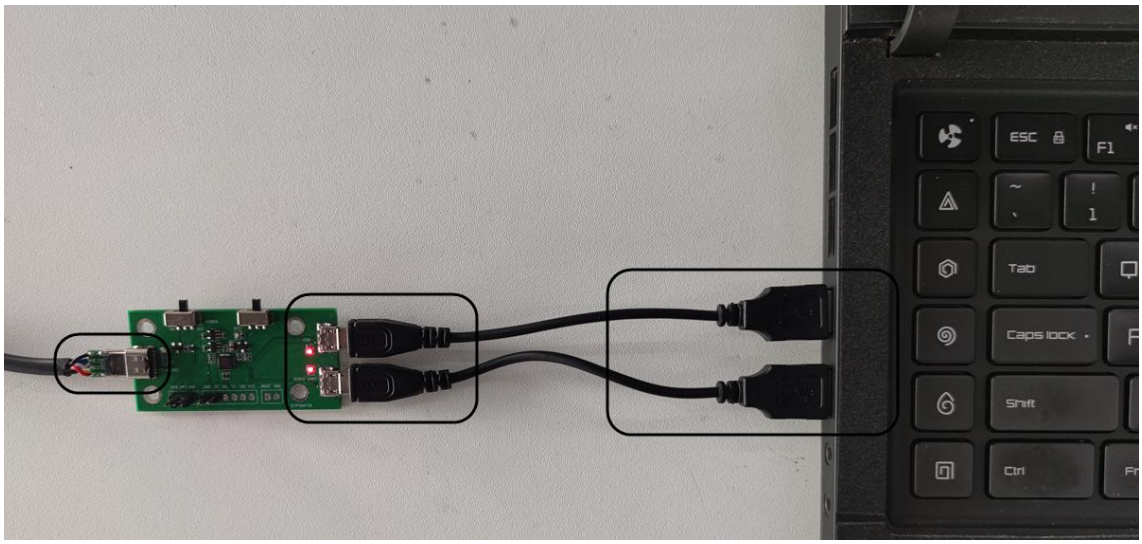


Figure 8.1 serial port module connection to PC

8.2 Serial port driver installation

Open the PC device manager(figure 8.2) and select the update drive software on the *USB Serial* device (the drive file directory recorded in Chapter 6.1), then complete the serial port module drive installation(figure 8.3/8.4/8.5).

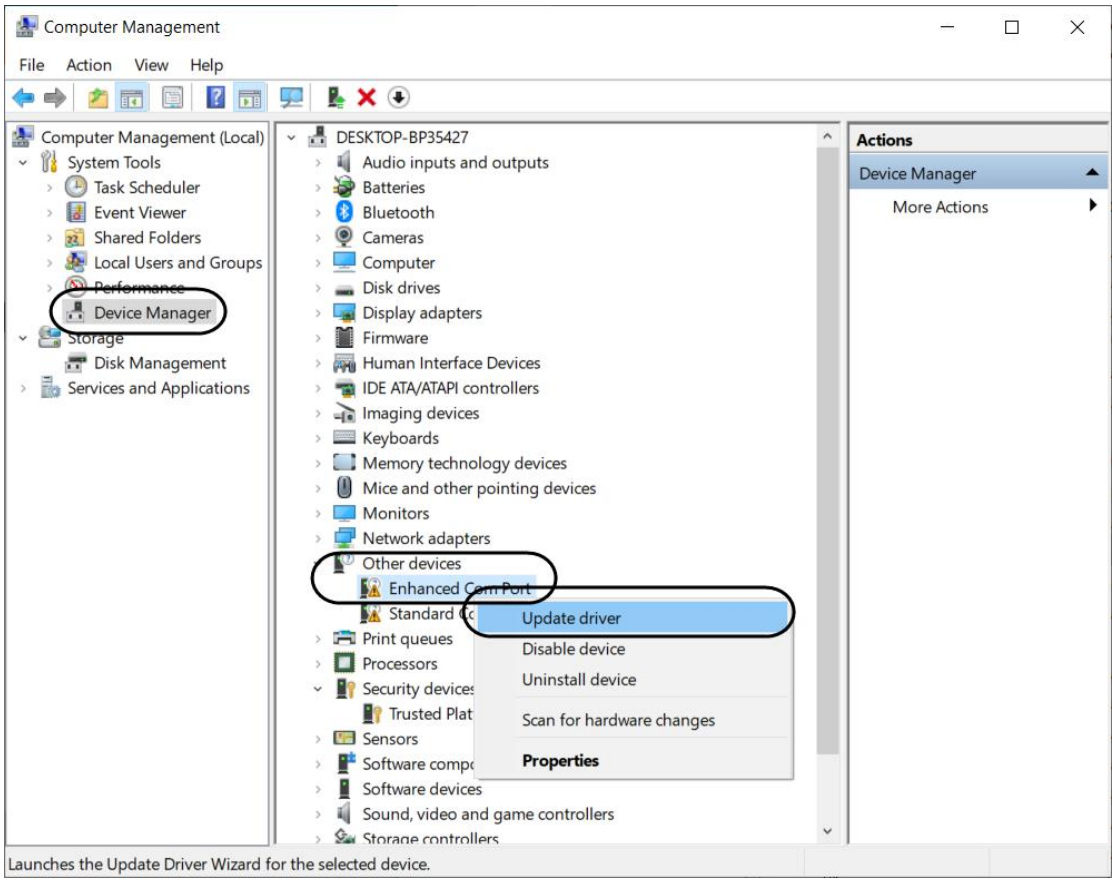


Figure 8.2 serial port drive uninstalled

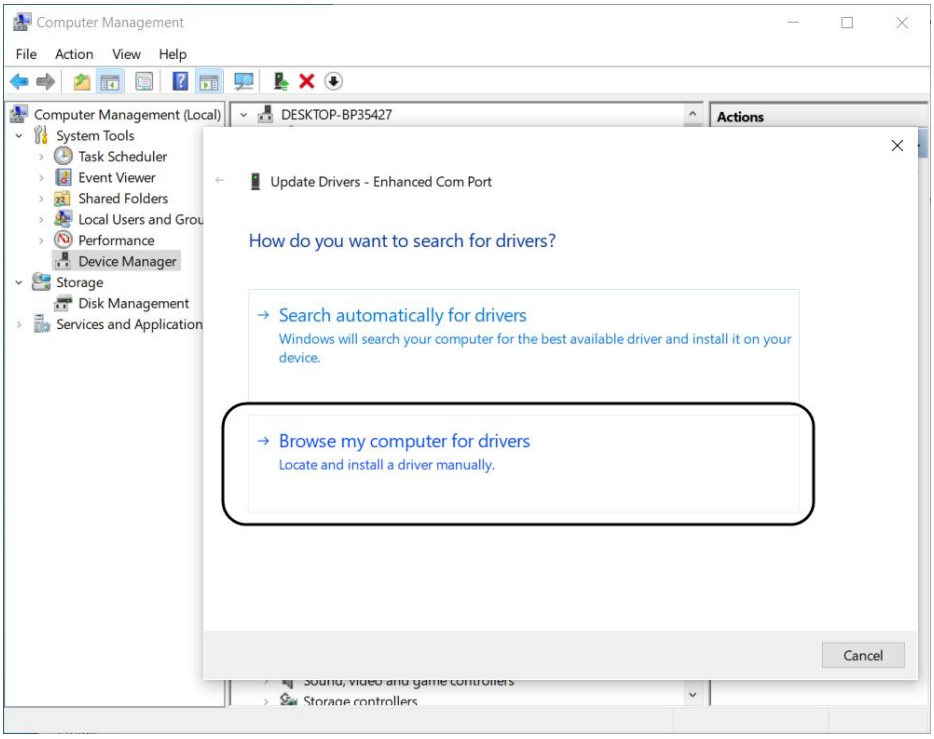


Figure 8.3 browse and select the drive directory

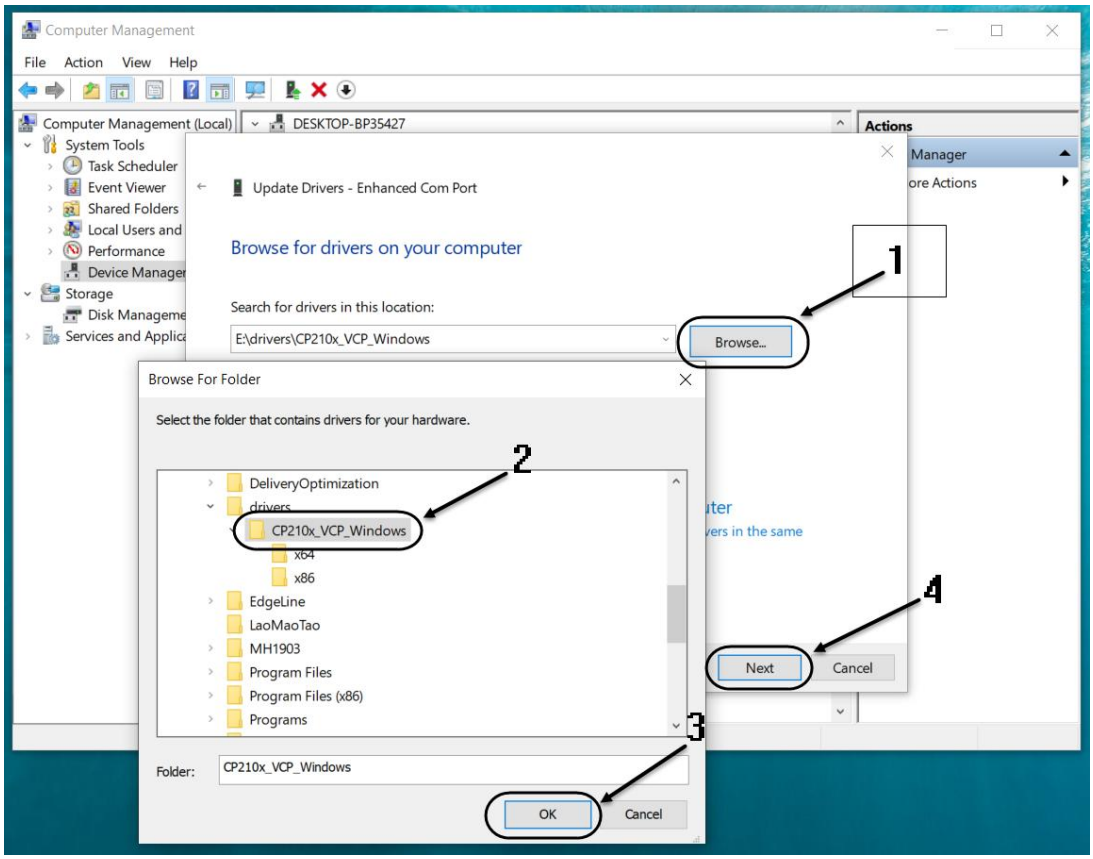


Figure 8.4 select the drive path of the serial port module

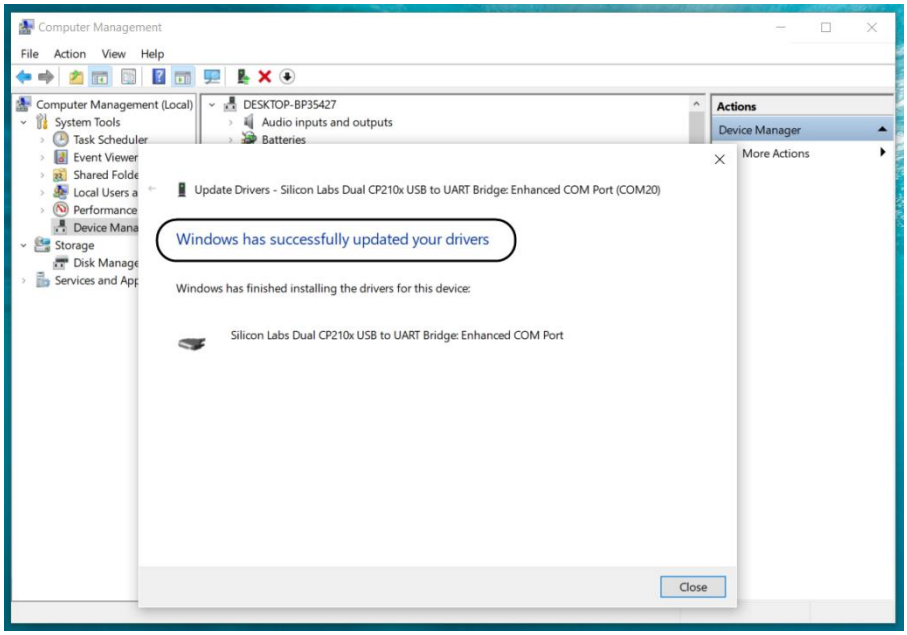


Figure 8.5 serial port module driver installed

Disconnect the connection between the PC computer and the serial port module, and reconnect the cable of the serial port module. The PC computer will automatically install the drives for the other serial ports After the drive

installation is completed, four sets of serial ports will appear in the device manager and the first set number of serial ports will be recorded(figure 8.6).

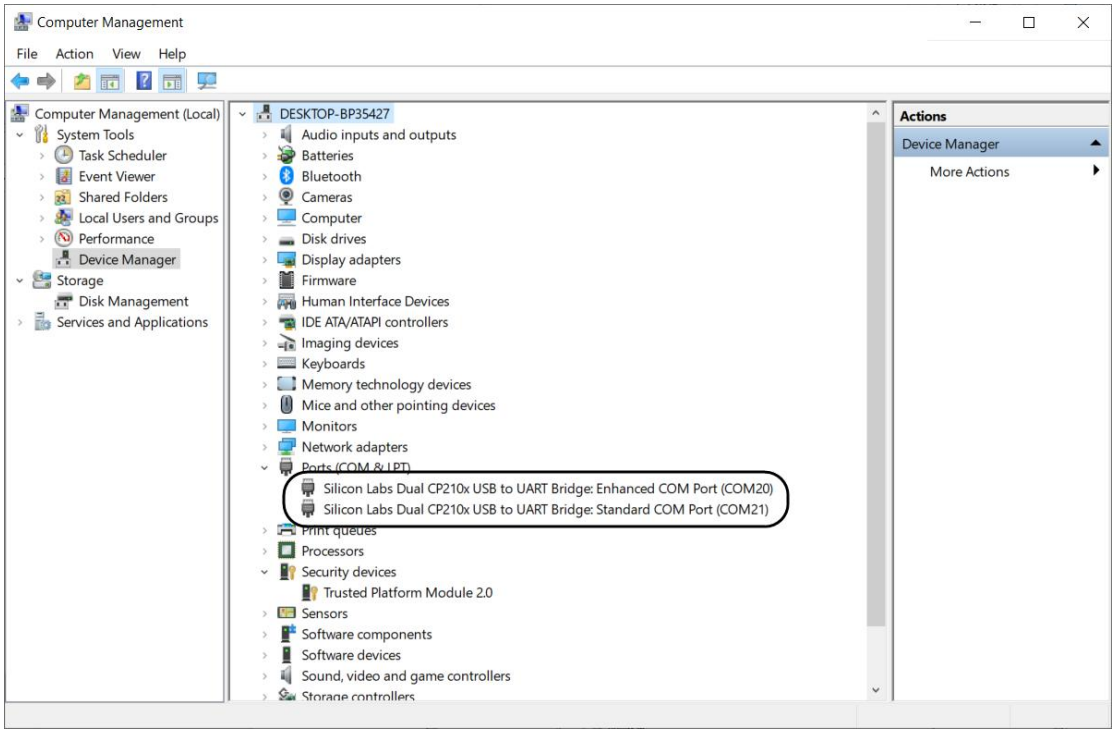


Figure 8.6 four serial ports

8.3 CoolWatcher

It is recommended that the computer configuration will not be less than 4G memory and dual-core processor, and the corresponding serial port drive must be installed to ensure that the 8910DM can connect and communicate with the PC. The supportable PC system version is Windows XP/7/10.

Double-click the Coolwatcher.exe file in the software directory to start the software. The configuration window goes active with model selection on the left and items on the right after Coolwatcher startup(figure 8.7).

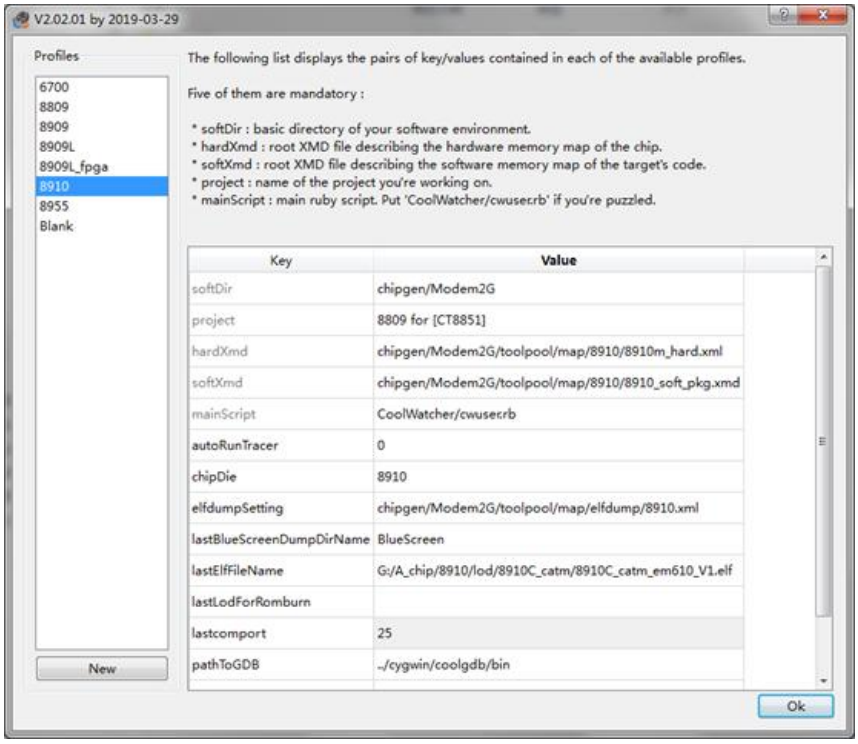


Figure 8.7 Coolwatcher configuration window

Select “Profiles” and “8910”, modify the necessary configuration items, such as port number, and fill in the serial port number recorded in Chapter 6.2.

Click the "OK" button, then the "Coolhost" configuration interface will be opened(figure 8.8) . Select the corresponding COM port, and click "connect".

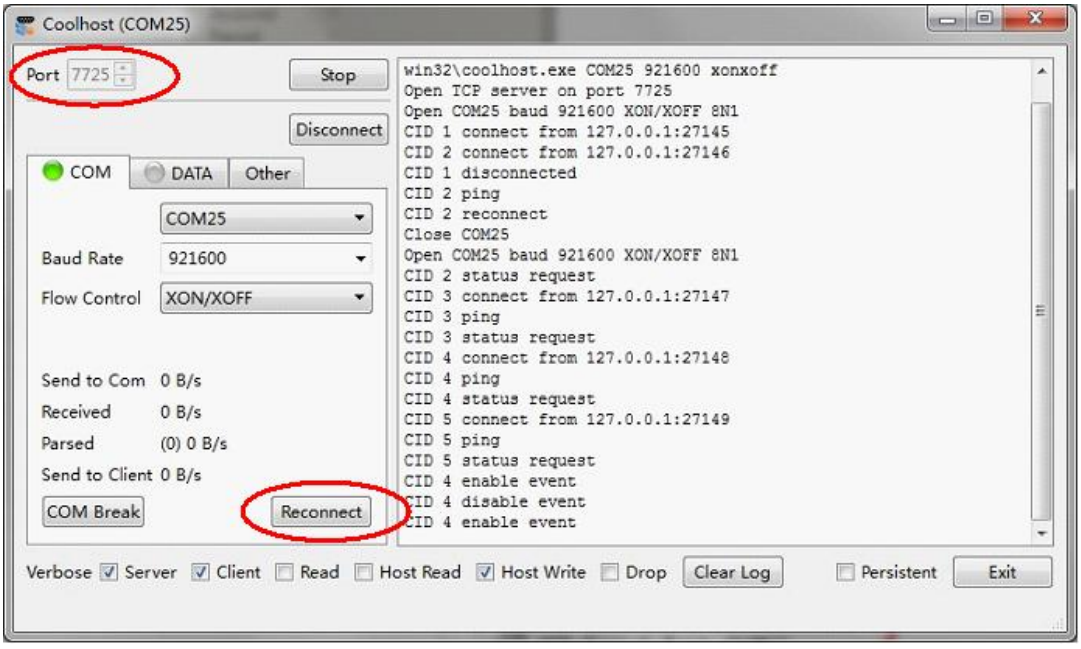


Figure 8.8 Coolwatcher connection

The main window of Coolwatcher includes: menu bar, tool bar, log print area, HW Library display tree, SW Library display tree, register view area, buffer read area and other functional areas. The tool can realize the functions of burning flash, checking buffer information, reading registers, viewing trace, dump data etc,.

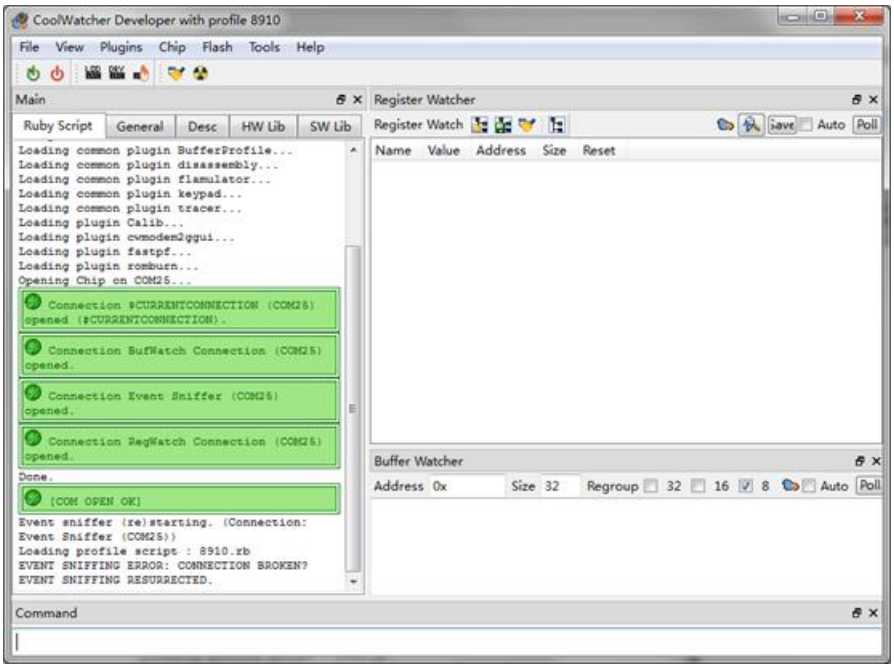


Figure 8.9 Coolwatcher desktop

After correct connection between the serial port module and the debug serial port pins of the device, click "Plugins"→"Activate Tracer" to open the Tracer log output interface (figure 8.10). Click the play button, and the tool will start to receive the Log output from the device and display it in scroll (figure 8.11).

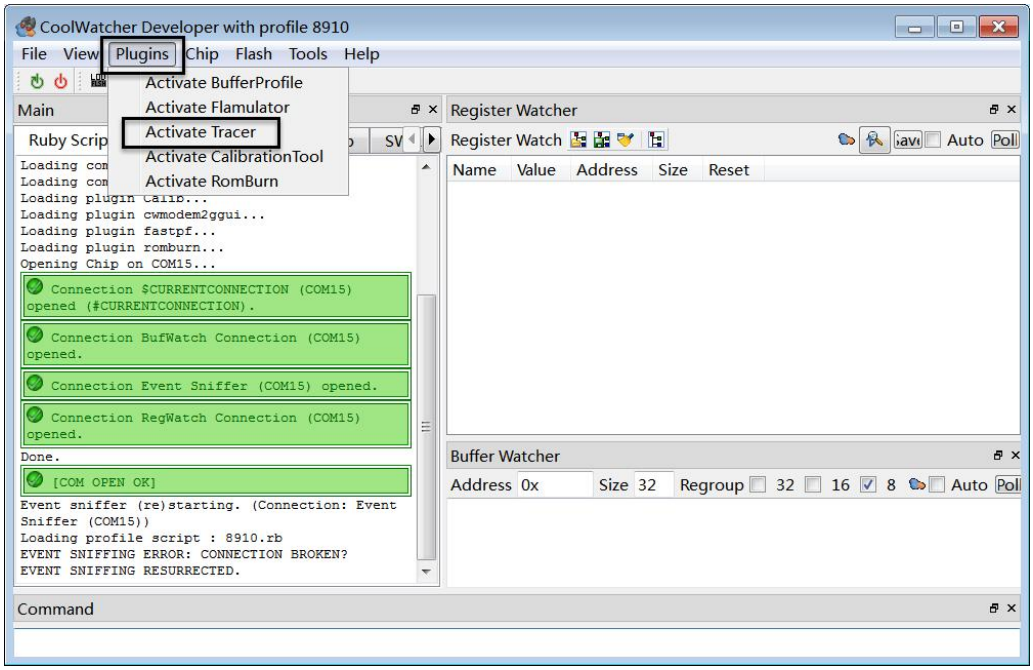


Figure8.10 open *activate tracer*

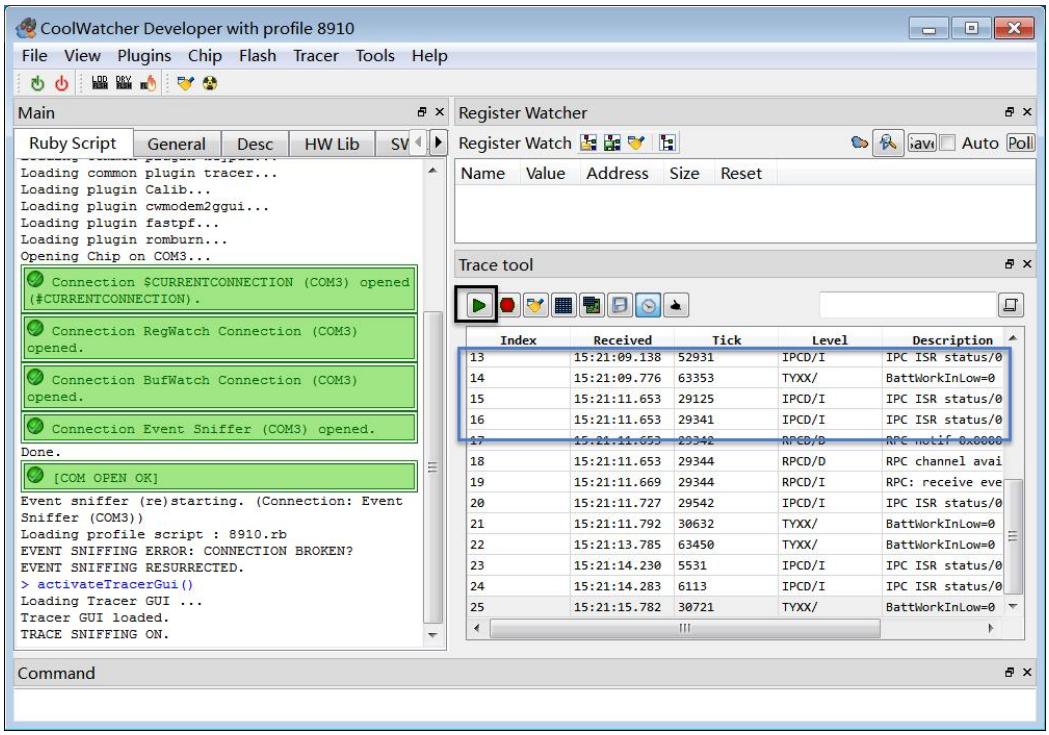


Figure 8.11 Coolwatcher Log

Click the ‘Clean Trace Windows’ button, and Coolwatch's Tracetool will clear the display of Log, as shown in Figure 8.12.

Click the ‘stop sniffing traces’ button, and the Coolwatcher stops receiving log data and ends recording log files, as shown in Figure 8.13.

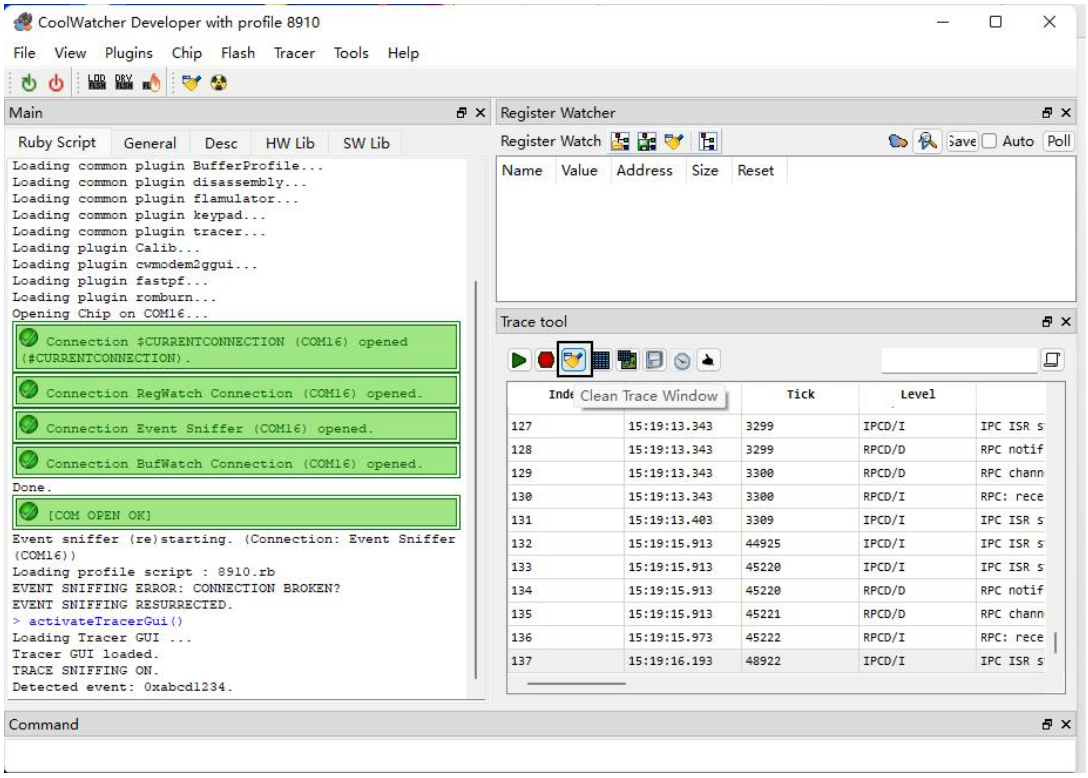


Figure 8.12 Clear the displayed log

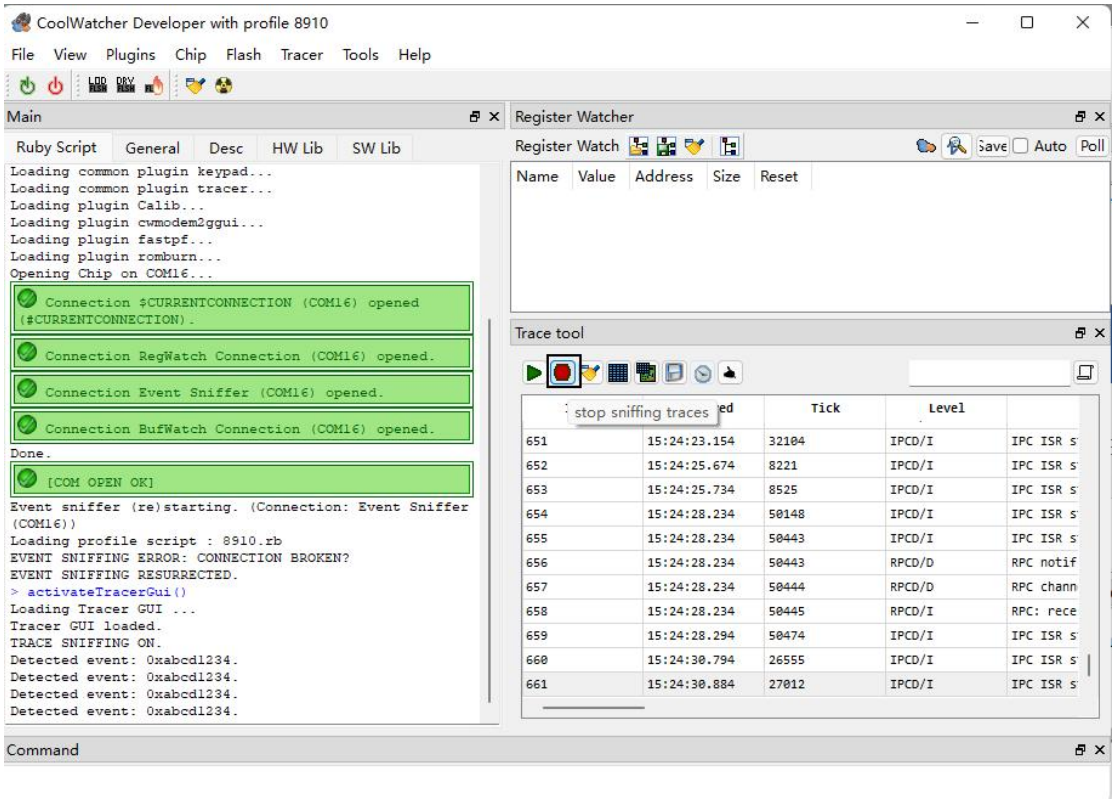


Figure 8.13 Stop Log Logging

For more configuration of tool use, please refer to the document 'Coolwatcher_User_Guide_EN.pdf'.

8.4 Output application debug log

DemoApp can add debugging information output to assist application development and debugging. Please use XX_debug_build.bat' compilation script in 'app_csdk' directory, this compilation script enables the Log output function by default, as shown in Figure 5.1.

Application add debug information, use the 'TRACE' interface, and the method of use similar to the 'printf()' format output function. The examples is provided in ‘main. C’ in ‘DemoApp’, as shown in Figure 8.14. Compile and generate the img firmware file, download it into the device, and the added log will be displayed on the coolwatcher tool, as shown in Figure 8.15.

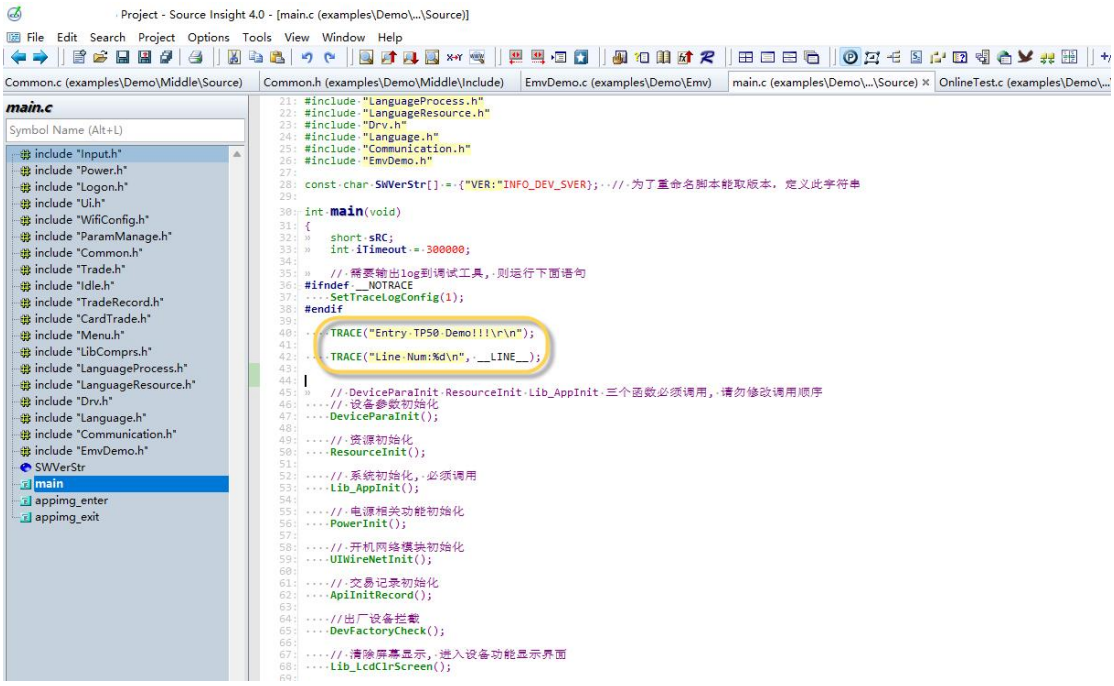


Figure 8.14 Add debug log

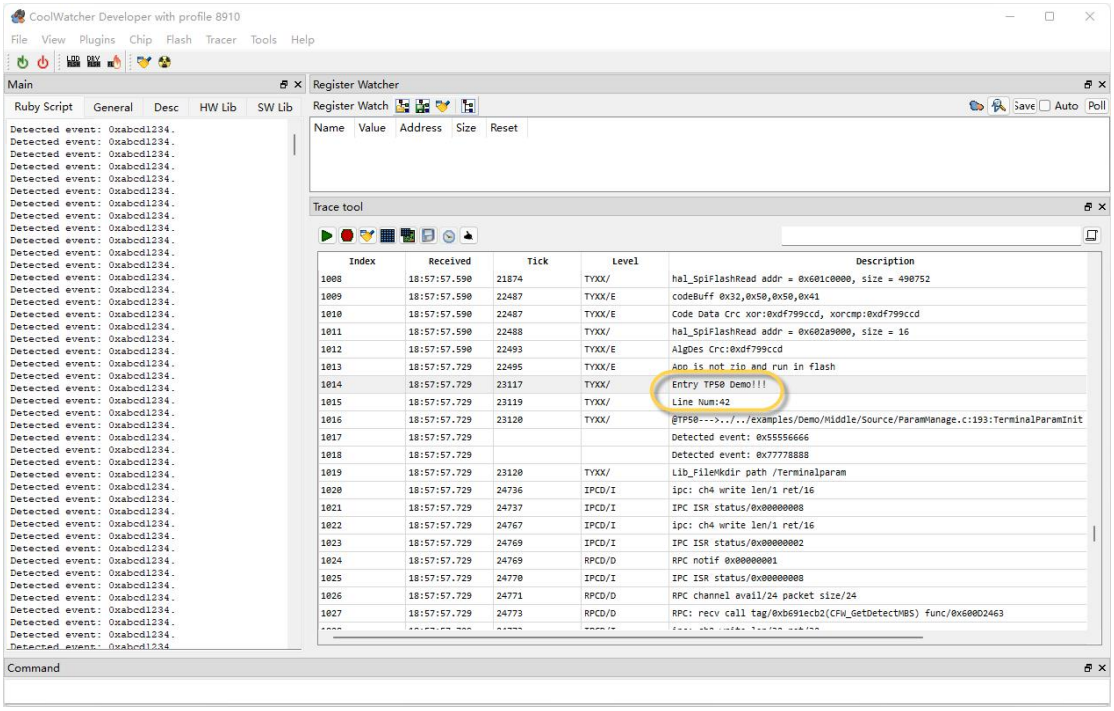


Figure 8.15 Show debug log

9 FS file update

The files in the device can be read, written and deleted through the FlashTool tool.

9.1 Open Flash Tool

The device first enters Maintenance mode and the operation steps are as follows:Main Menu->More->Maintenance. Open the ‘tools.exe' tool and click on the 'Device name' drop-down box，Click ‘Refresh List’ to refresh the device file system files list. As shown in Figure 9.1.

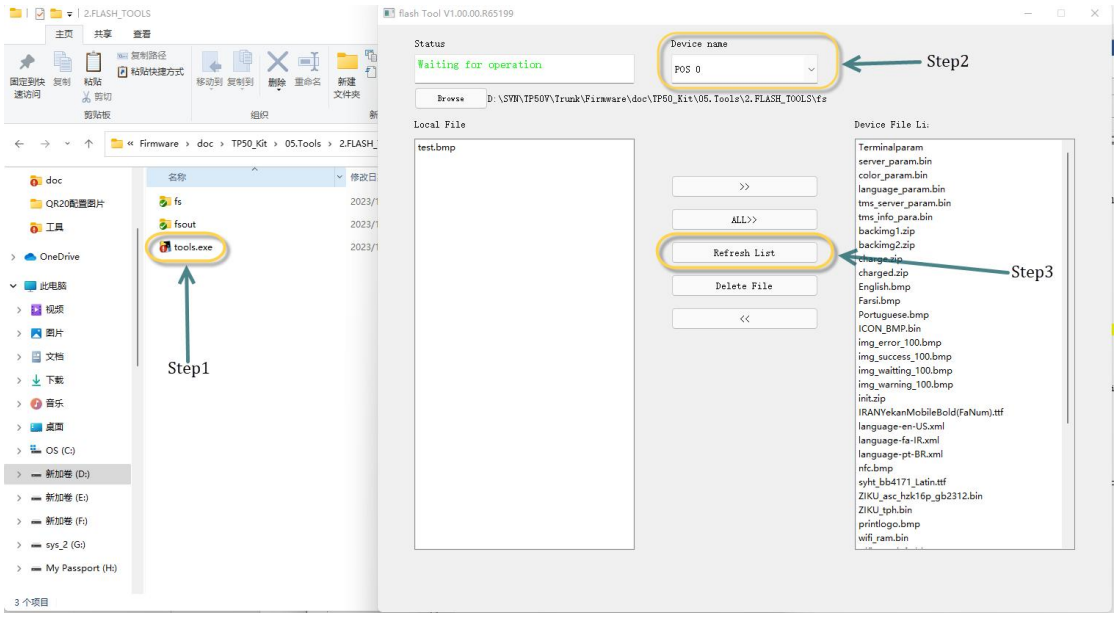


Figure 9.1 Open Flash Tool

9.2 Update file

Update the file in the device, first put the target file into the fs directory under the flash tool directory, select the file to be updated in the Local File area, and click ‘>>’ to write the selected file to the device. Click ‘All>>’ ,will write all files in the Local File area to the device(figure 9.2).

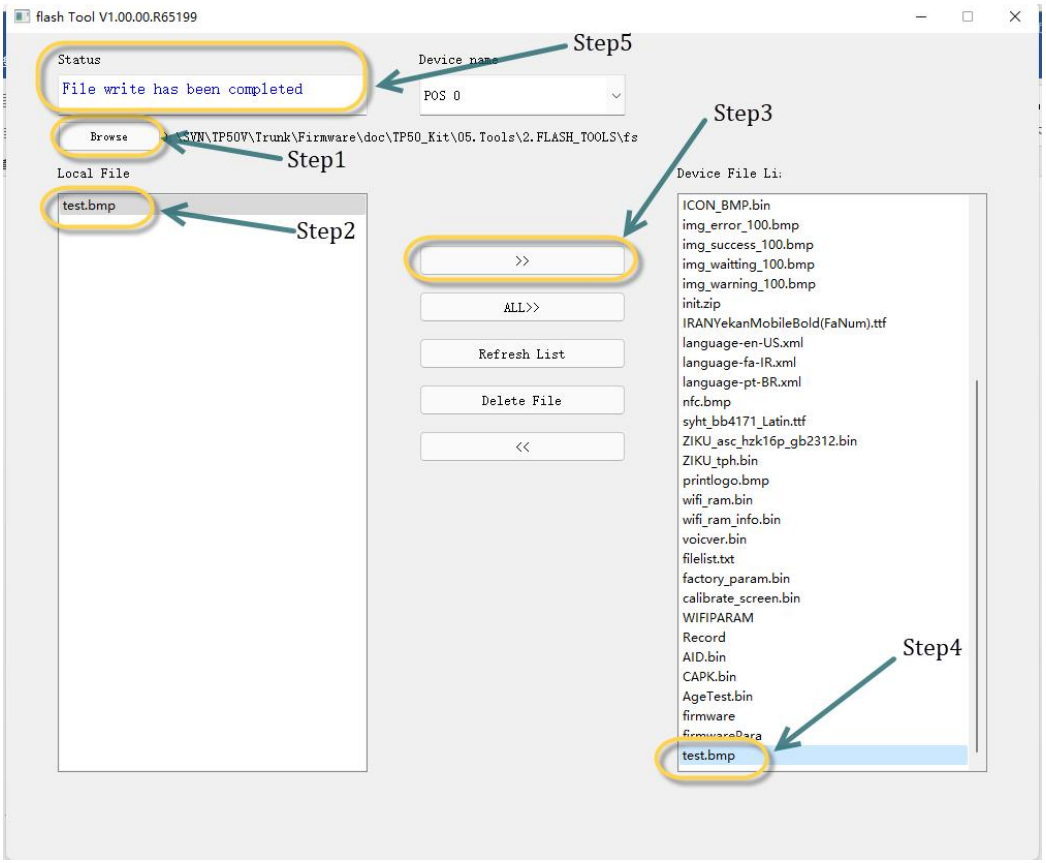


Figure 9.2 Update file to device

9.3 Read file

Read the file in the device, select the file to be read in the ‘Device File List’ area, and click ‘<<’. The selected file will be read from the device and stored in the ‘fsout’ directory under the Flash Tool directory(figure 9.3).

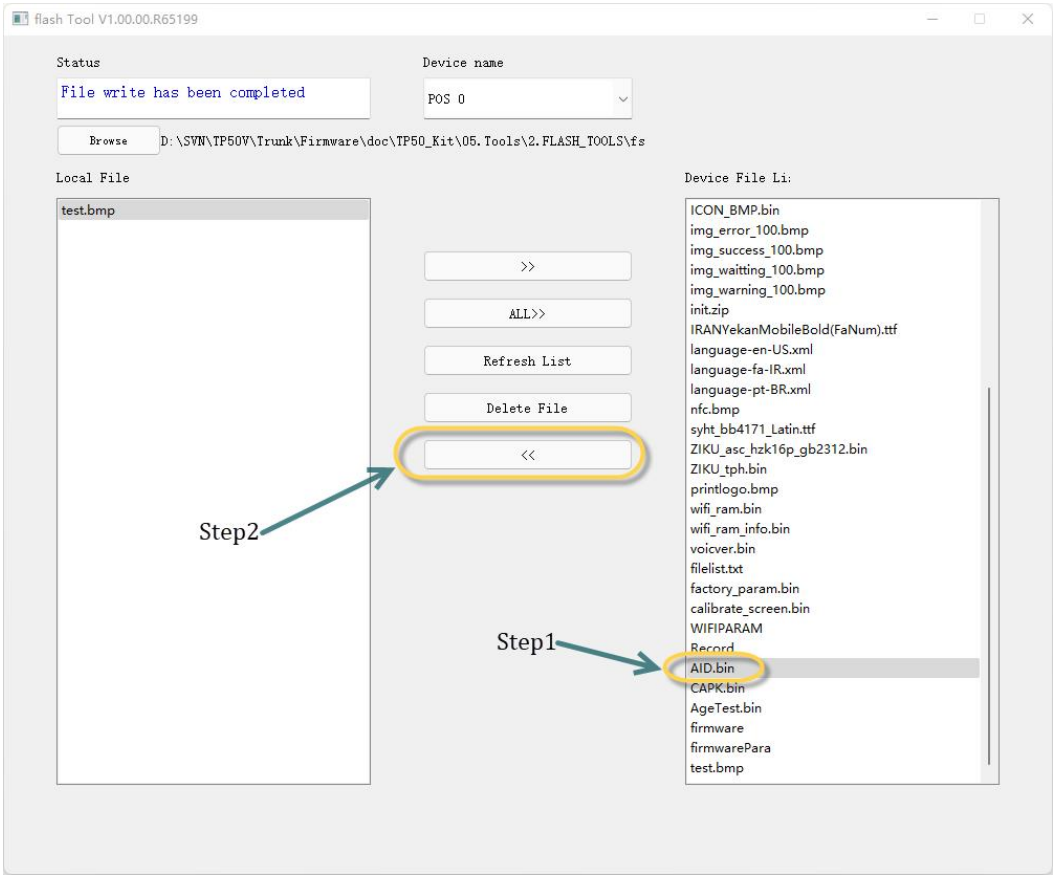


Figure 9.3 Read file from device

9.4 Delete file

Delete the file in the device, select the file to be deleted in the ‘Device File List’ area, click ‘Delete File’, and the selected file will be deleted from the device(figure 9.4).

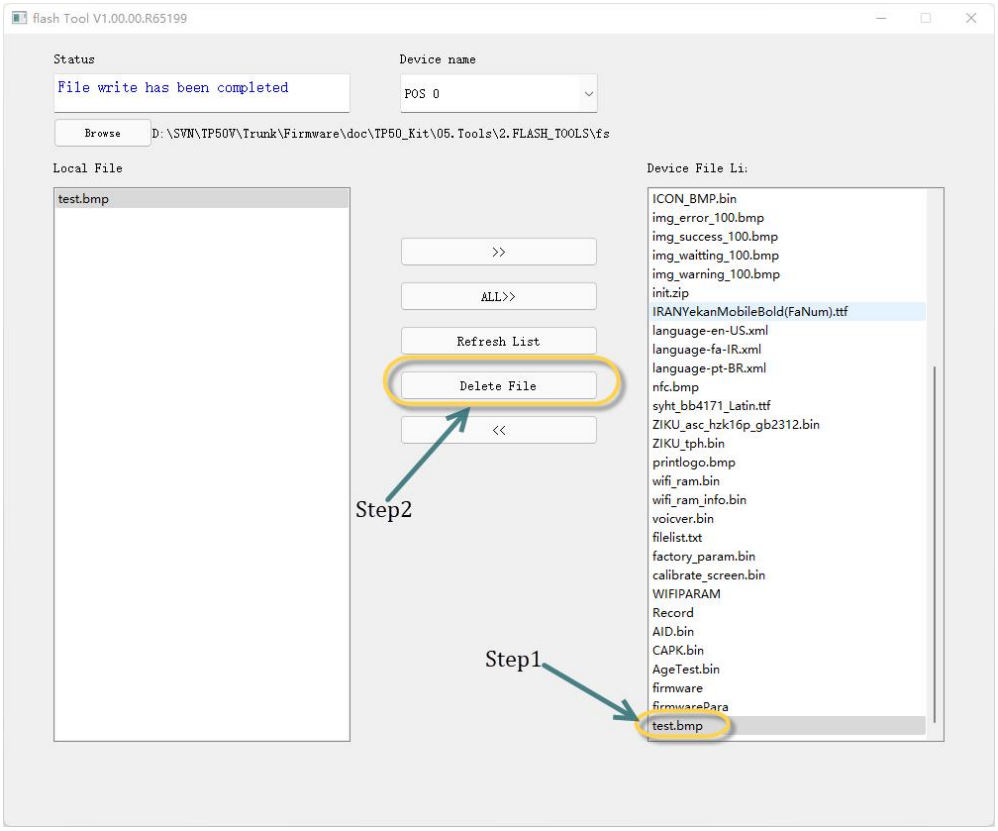


Figure 9.4 Delete file in device

10 Appendix

1. 8910_module_usb_driver_20191011_signed.7z is the main MCU driver package, provided separately;
2. prebuilts.rar is the SDK development environment compilation tool chain package, provided separately;
3. UpgradeDownload.exe is a firmware update tool, provided separately;
4. CP210x_VCP_Windows.rar is the driver software package for the USB virtual serial port enulation, provided separately.